

```
student = {
    "Student_ID": 201,
    "Student_Name": "Vismithaa",
    "Department": "AI and ML"
}

company = {
    "Company_ID": 1,
    "Company_Name": "TCS",
    "Location": "Chennai"
}

mentor = {
    "Mentor_ID": 101,
    "Mentor_Name": "Dr. Ramesh",
    "Designation": "Assistant Professor"
}

internship = {
    "Internship_ID": 301,
    "Internship_Title": "ML Intern",
    "Internship_Status": "Pending",
    "Company_ID": 1,
    "Mentor_ID": 101,
    "Read_TS": 0,
    "Write_TS": 0
}

evaluation = {
    "Student_ID": 201,
    "Internship_ID": 301,
    "Marks": 95,
    "Grade": "A"
}

transactions = {
    "T1": 10,
    "T2": 20,
    "T3": 5
}

print("===== INITIAL DATABASE =====\n")
print("\nInternship Table")
```

```

print(internship)

def read_operation(transaction):
    ts = transactions[transaction]
    print("\n-----")
    print(transaction, "attempts READ operation")
    print("-----")
    if ts < internship["Write_TS"]:
        print("READ REJECTED")
        print(f"Reason : TS({transaction}) = {ts} < Write_TS = {internship['Write_TS']}")
    else:
        internship["Read_TS"] = max(internship["Read_TS"], ts)
        print("READ SUCCESSFUL")
        print("Student :", student["Student_Name"])
        print("Internship :", internship["Internship_Title"])
        print("Current Status :", internship["Internship_Status"])
        print("Read_TS Updated To :", internship["Read_TS"])

def write_operation(transaction, new_status):
    ts = transactions[transaction]
    print("\n-----")
    print(transaction, "attempts WRITE operation")
    print("-----")
    if ts < internship["Read_TS"]:
        print("WRITE REJECTED")
        print(f"Reason : TS({transaction}) = {ts} < Read_TS = {internship['Read_TS']}")
        return
    if ts < internship["Write_TS"]:
        print("WRITE REJECTED")
        print(f"Reason : TS({transaction}) = {ts} < Write_TS = {internship['Write_TS']}")
        return
    internship["Internship_Status"] = new_status
    internship["Write_TS"] = ts
    print("WRITE SUCCESSFUL")
    print("Updated Internship Status :", internship["Internship_Status"])
    print("Write_TS Updated To :", internship["Write_TS"])

read_operation("T1")
write_operation("T1", "Completed")
read_operation("T2")
write_operation("T2", "Completed")
write_operation("T3", "Pending")

print("\n===== FINAL DATABASE =====\n")
print("Student Table")

```

```
print(student)
print("\nCompany Table")
print(company)
print("\nMentor Table")
print(mentor)
print("\nInternship Table")
print(internship)
print("\nEvaluation Table")
print(evaluation)
```

```
Internship Table
{'Internship_ID': 301, 'Internship_Title': 'ML Intern', 'Internship_Status': 'Pending', 'Company_ID': 1, 'Mentor_ID': 101, 'Read_TS': 0, 'Write_TS': 0}
```

```
-----
T1 attempts READ operation
-----
```

```
READ SUCCESSFUL
Student : Vismitaa
Internship : ML Intern
Current Status : Pending
Read_TS Updated To : 10
```

```
-----
T1 attempts WRITE operation
-----
```

```
WRITE SUCCESSFUL
Updated Internship Status : Completed
Write_TS Updated To : 10
```

```
-----
T2 attempts READ operation
-----
```

```
READ SUCCESSFUL
Student : Vismitaa
Internship : ML Intern
Current Status : Completed
Read_TS Updated To : 20
```

```
-----
T2 attempts WRITE operation
-----
```

```
WRITE SUCCESSFUL
Updated Internship Status : Completed
Write_TS Updated To : 20
```

```
WRITE REJECTED
```

```
Reason : TS(T3) = 5 < Read_TS = 20
```

```
===== FINAL DATABASE =====
```

```
Student Table
```

```
{'Student_ID': 201, 'Student_Name': 'Vismitaa', 'Department': 'AI and ML'}
```

```
Company Table
```

```
{'Company_ID': 1, 'Company_Name': 'TCS', 'Location': 'Chennai'}
```

```
Mentor Table
```

```
{'Mentor_ID': 101, 'Mentor_Name': 'Dr. Ramesh', 'Designation': 'Assistant Professor'}
```

```
Internship Table
```

```
{'Internship_ID': 301, 'Internship_Title': 'ML Intern', 'Internship_Status': 'Completed', 'Company_ID': 1, 'Mentor_ID': 101, 'Read_TS': 20, 'Writ
```

```
Evaluation Table
```

```
{'Student_ID': 201, 'Internship_ID': 301, 'Marks': 95, 'Grade': 'A'}
```